

# Ernie Chu

PhD Student

Department of Computer Science, Johns Hopkins University

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## Education

### Johns Hopkins University

PhD in Computer Science

- Advised by Dr. Vishal M. Patel

Baltimore, MD

Sept. 2024 - PRESENT

### Johns Hopkins University

M.S.E. in Computer Science

Baltimore, MD

Sept. 2024 - Aug. 2026

### National Sun Yat-sen University (NSYSU)

B.S. in Computer Science and Engineering

- GPA 4.15/4.3
- Dean's list awards
- NSTC Research Grant for University Students

Kaohsiung, Taiwan

Sept. 2018 - June 2022

## Experience

### Pinterest, Inc.

Machine Learning Engineer Intern

Palo Alto, CA

May. 2026 - Aug. 2026

- Designed and implemented a token-span GRPO algorithm for Pinterest's in-house image captioner, improving object accuracy from 93% to 96% as evaluated by an LLM-as-a-judge pipeline.
- Redesigned the caption output schema into a more verifiable format, enabling automated token-level reward assignment and robust feedback loops for future RL training pipelines.

### Academia Sinica

Research Assistant. Advisor: Jun-Cheng Chen

Taipei, Taiwan

Sept. 2022 - Aug. 2024

- Video generations using image Diffusion Models
- Generalizable facial video forgery (Deepfake) detection
- Adversarial attacks, generative models, multi-modal reasoning

### Office of International Affairs, NSYSU

Full-Stack Web Developer (Part-Time)

Kaohsiung, Taiwan

Sept. 2018 - June 2022

- New online system from scratch for the exchange program
- Maintain all websites under the office
- HTML/JS/CSS, PHP, Vue.js, Express.js, Sequelize.js, MariaDB, MongoDB

## Selected Publications & Preprints

- [1] **Ernie Chu**, Vishal Patel, **Not All Tokens Need 40 Steps: Heterogeneous Step Allocation in Diffusion Transformers for Efficient Video Generation**. *arXiv preprint*, 2026. 
- [2] **Ernie Chu**, I-Sheng Fang, Tai-Ming Huang, Pin-Yen Chiu, Vishal Patel, Jun-Cheng Chen, **Adapt2Hide: Leveraging Off-The-Shelf Autoencoder for Reversible Visual Processing**. *ICIP*, 2026 (oral, top 3%). 
- [3] Yang-Che Sun, Cheng Yu Yeo, **Ernie Chu**, Jun-Cheng Chen, Yu-Lun Liu, **FIPER: Factorized Features for Robust Image Super-Resolution and Compression**. *NeurIPS*, 2025. 
- [4] Shuo-Yen Lin, **Ernie Chu**, Po-Hung Yeh, Jun-Cheng Chen, Jia-Ching Wang, **Diffusion to Confusion: Naturalistic Adversarial Patch Generation Based on Diffusion Model for Object Detector**. *ICIP*, 2025. 
- [5] ChunYen Shih, Li-Xuan Peng, Jia-Wei Liao, **Ernie Chu**, Cheng-Fu Chou, Jun-Cheng Chen, **Pixel Is Not A Barrier: An Effective Evasion Attack for Pixel-Domain Diffusion Models**. *AAAI*, 2025. 
- [6] Tai-Ming Huang, Yue-Hua Han, **Ernie Chu**, Shu-Tzu Lo, Kai-Lung Hua, and Jun-Cheng Chen, **Generalized Image-based Deepfake Detection through Foundation Model Adaptation**. *ICPR*, 2024. 
- [7] **Ernie Chu**, Tzuhsuan Huang, Shuo-Yen Lin, and Jun-Cheng Chen, **MeDM: Mediating Image Diffusion Models for Video-to-Video Translation with Temporal Correspondence Guidance**. *AAAI*, 2024. 
- [8] Po-Han Huang, Yue-Hua Han, **Ernie Chu**, Jun-Cheng Chen, and Kai-Lung Hua, **Multi-Task Self-Blended Images for Face Forgery Detection**. *ACM Multimedia Asia*, 2023. 

## Professional Services

2024 **Reviewer**, The Thirteenth International Conference on Learning Representations (ICLR 2025)

Singapore

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|------|------------------------------------------------------------------------------------------------|------------------|
| 2023 | <b>Reviewer</b> , The Twelfth International Conference on Learning Representations (ICLR 2024) | Vienna, Austria  |
| 2023 | <b>Reviewer</b> , 18th International Conference on Machine Vision Applications (MVA 2023)      | Hamamatsu, Japan |

## Honors

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|------|--------------------------------------------------------------------------------------------|--------|
| 2022 | <b>Honorary Member</b> , The Phi Tau Phi Scholastic Honor Society of the Republic of China | Taiwan |
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## Projects

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|------------------------------------------------------------------------|-----------------------|
| <b>Interactive transcriptor for 2024 Taiwan presidential campaigns</b> | Kaohsiung, Taiwan     |
| Leading Developer                                                      | Oct. 2023 - Nov. 2023 |

- An interactive transcriptor referencing to the source videos of 2024 Taiwan (ROC) presidential campaigns in campus
- Helping students digest 1-hour presentation in 10 minutes, reinforcing the democratic resilience in campus  

|                   |                      |
|-------------------|----------------------|
| <b>LoLViz.</b>    | Kaohsiung, Taiwan    |
| Leading Developer | May 2022 - June 2022 |

- A 3-stage League of Legends visualizer
- Vue.js, D3.js, Firebase   

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|-----------------------------|-----------------------|
| <b>NSYSU Captcha Solver</b> | Kaohsiung, Taiwan     |
| Leading Developer           | Jan. 2021 - Feb. 2021 |

- CNN for auto filling the captcha verification
- TensorFlow.js, Chrome extension 

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|----------------------------------------|-------------------|
| <b>Secure On-line Chatting Service</b> | Kaohsiung, Taiwan |
| Leading Developer                      | Spring 2021       |

- On-line chatting service with console-based UI using ncurses
- Socket programming, multi-threading C++17, attribute-based encryption   

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|------------------------------------------|-------------------|
| <b>License Plate Verification System</b> | Kaohsiung, Taiwan |
| Leading Developer                        | Fall 2020         |

- ML-based object detector to track and verify the scooter license plate, deployed on Nvidia Jetson Nano  